

Research Data Analysis

Gemini Dataset

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1st presentation session (9 Nov 2025)

Introduction

In this session, our main goal was to analyze the data and examine the relationships between variables (except *conversations* and *responses*) in the dataset. Using various statistical and data analysis methods, we attempted to identify the relationships between different variables (focusing on the *Conv_metadata* column) and examine the impact of each of them on the winning of the LLM model.

Introducing data and variables

The dataset used for this project includes various information such as *Conv_metadata* variables and models participating in the competition, *winner*; *conversations* of each model, *responses* of each model, *language*, and *category_tag*. These variables specifically included structural information of the conversations, and our goal was to examine the relationship of these variables with the Gemini model winning.

Methodology

To analyze the data, first the desired file was imported and then in the cleaning and extraction phase of the desired features, the columns that had only one value or were all *NaN* were deleted and the *winner_name* column was created to identify the winner, then the *Conv_metadata* column, since it has information in the form of a *Dict*, was flattened and the relevant information was extracted. Due to the large amount of data, only the English language was examined and then the desired columns were separated for analysis. Finally, using various methods, graphs including correlation analysis were examined to show the relationship between different variables.

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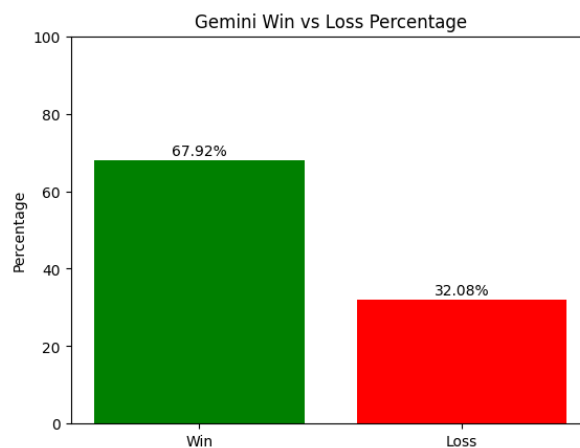
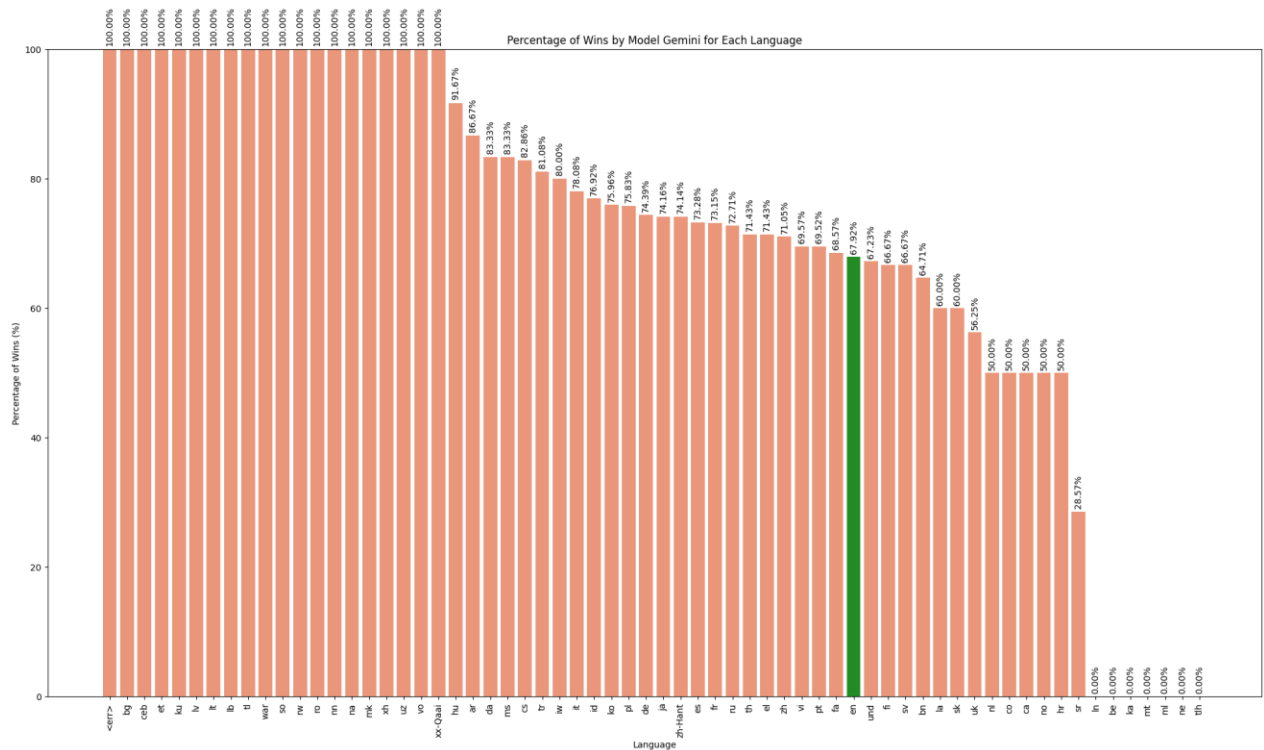
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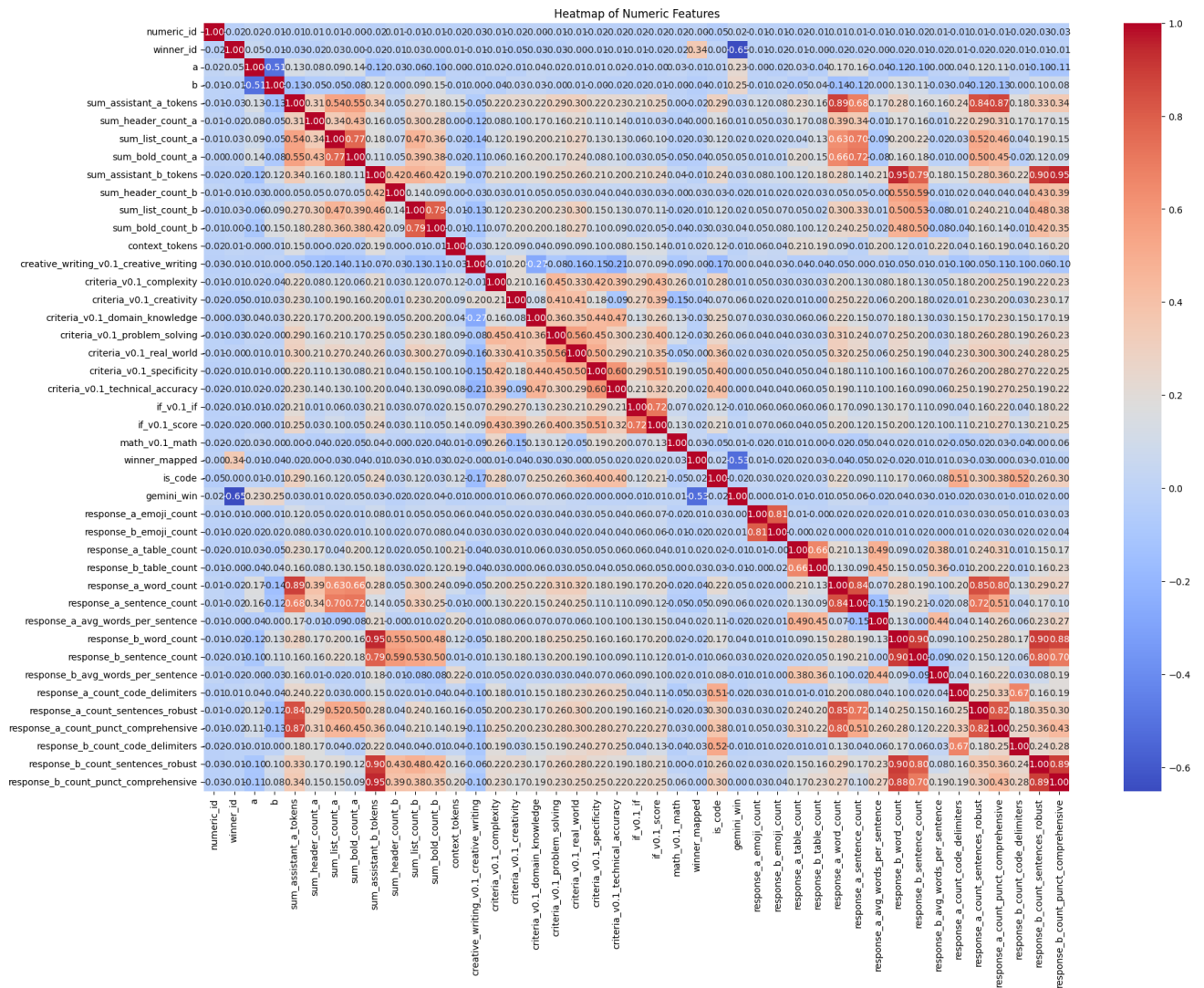
Analysis results

Plot 1: Gemini Win/Loss Analysis

In this step, we analyze the overall performance of the Gemini model in all languages by comparing its win and loss counts across all matches. We calculate the total number of matches, determine how many were wins or losses (including ties), compute the corresponding percentages, and visualize the results using a bar chart. This provides a quick overview of Gemini's general effectiveness in comparison to other models.

Second chart shows the winning and losing rates of the Gemini model in English language, and according to the figure, the winning rate of the Gemini model was almost 68% and the model's losing rate was 32%.



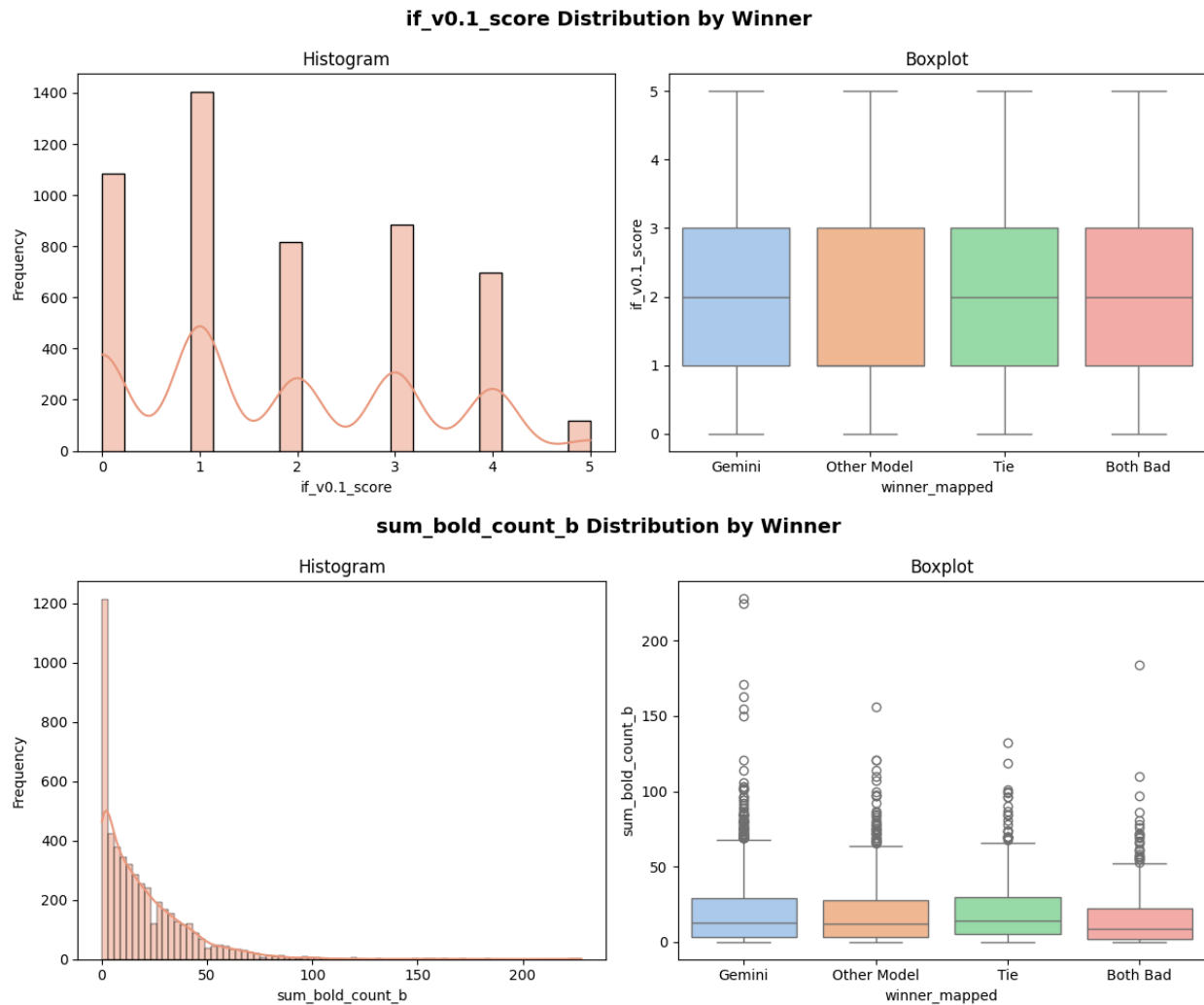


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Plot 3: Feature Distribution Exploration

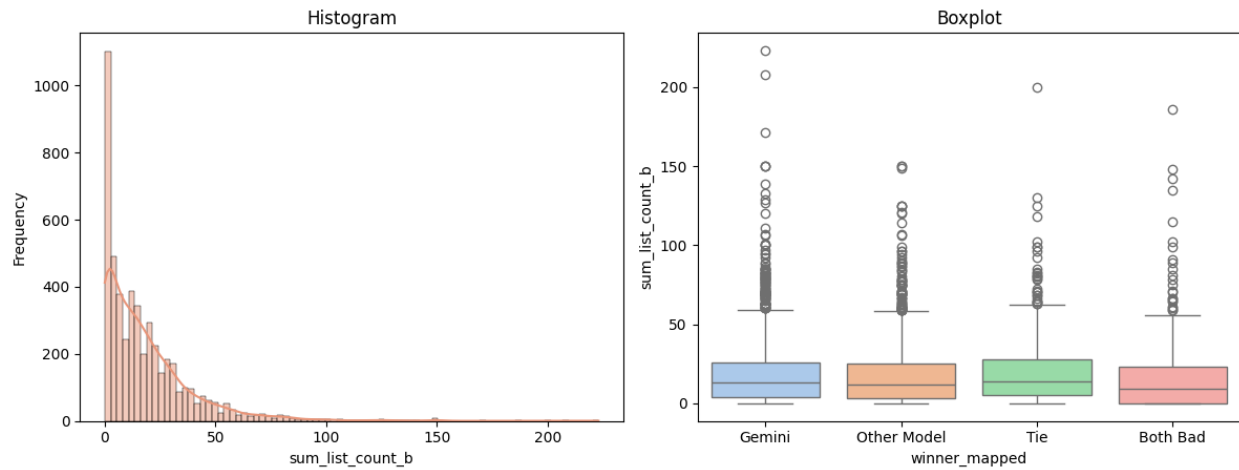
In this step, we visualize the overall distributions of key numerical features across the dataset using histograms and boxplots. The goal is to gain a general understanding of how each feature behaves (e.g., skewness, outliers, or spread). At this stage, the **winner model is not yet incorporated** into the analysis — the exploration remains broad and unspecific. In the next steps, we will refine these analyses by comparing the distributions for each winner category and model-related characteristics.



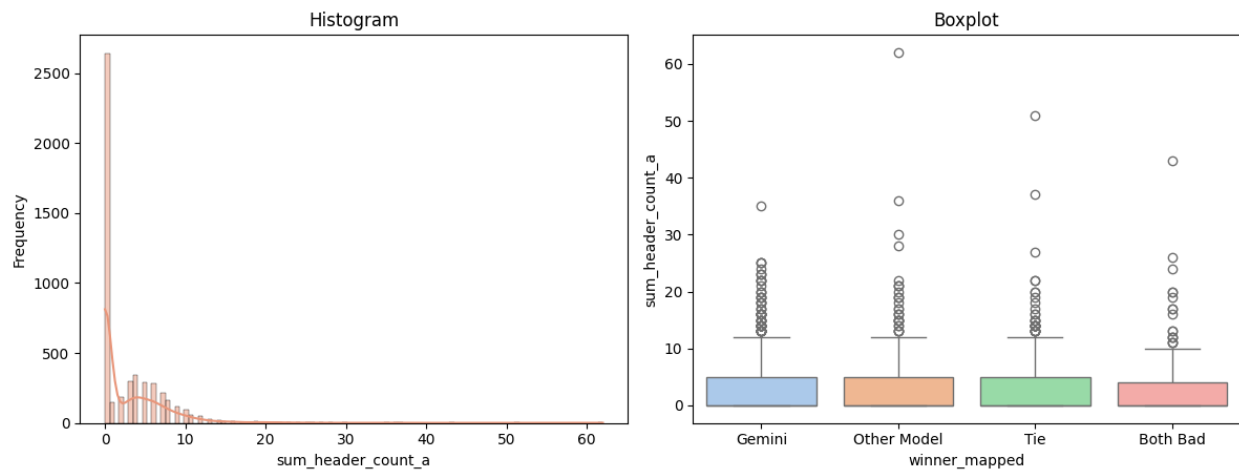
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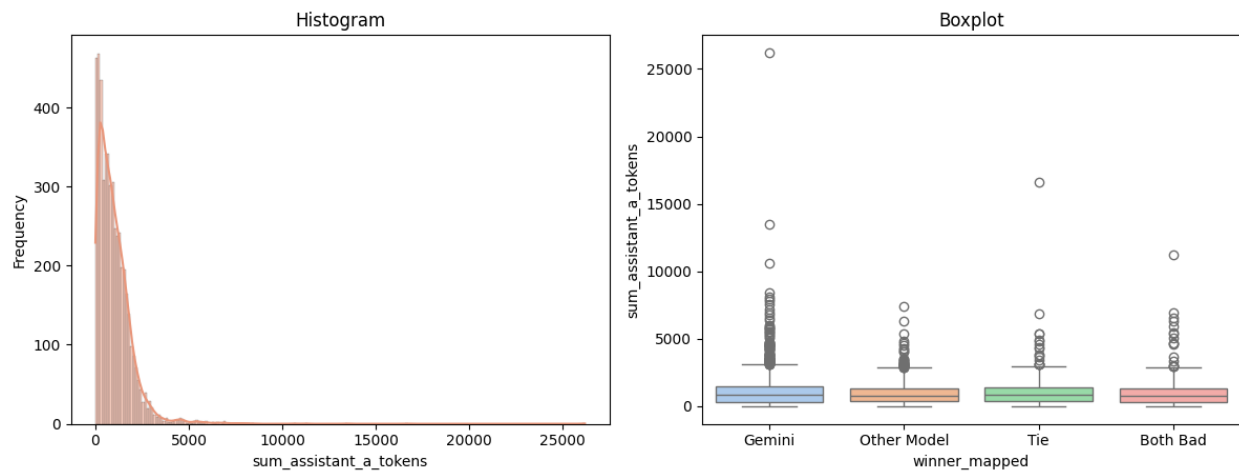
sum_list_count_b Distribution by Winner



sum_header_count_a Distribution by Winner



sum_assistant_a_tokens Distribution by Winner

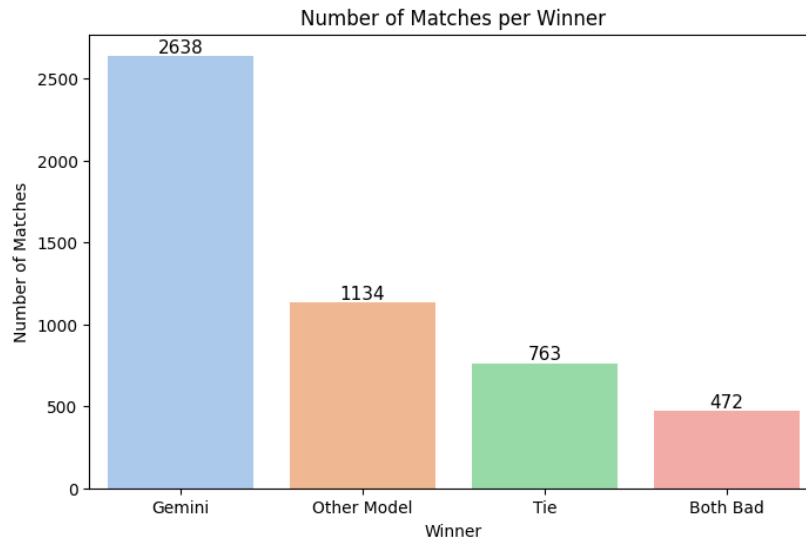


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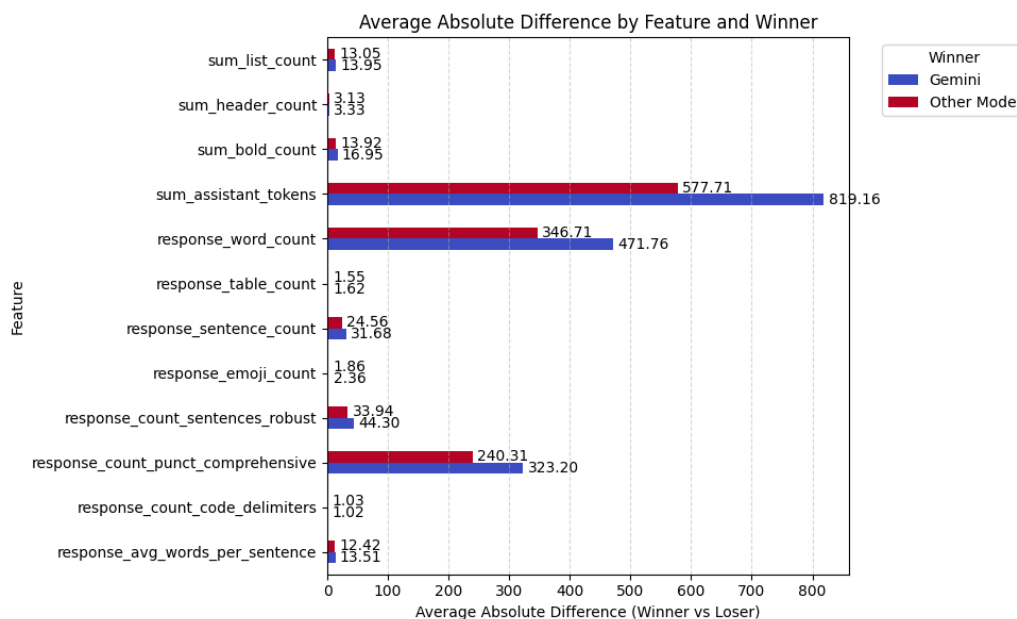
Plot 4: Winner Count Overview

In this step, we visualize the number of matches for each winner category: **Gemini**, **Other Model**, **Tie** and **Both Bad**. The bar chart shows the count of matches per outcome, providing an overview of how match results are distributed across different categories. This helps to quickly see which outcomes are more frequent and whether the dataset is balanced.



Plot 5: Average Absolute Difference Analysis

In this step, we analyze the differences between winning and losing models for selected features. We calculate the **absolute difference** between winner and loser values for each match, excluding ties and both_bad cases. Then, we compute the **average absolute difference** per feature and winner category and visualize it using a horizontal bar chart. This helps identify which features differ most between winners and losers across the dataset.

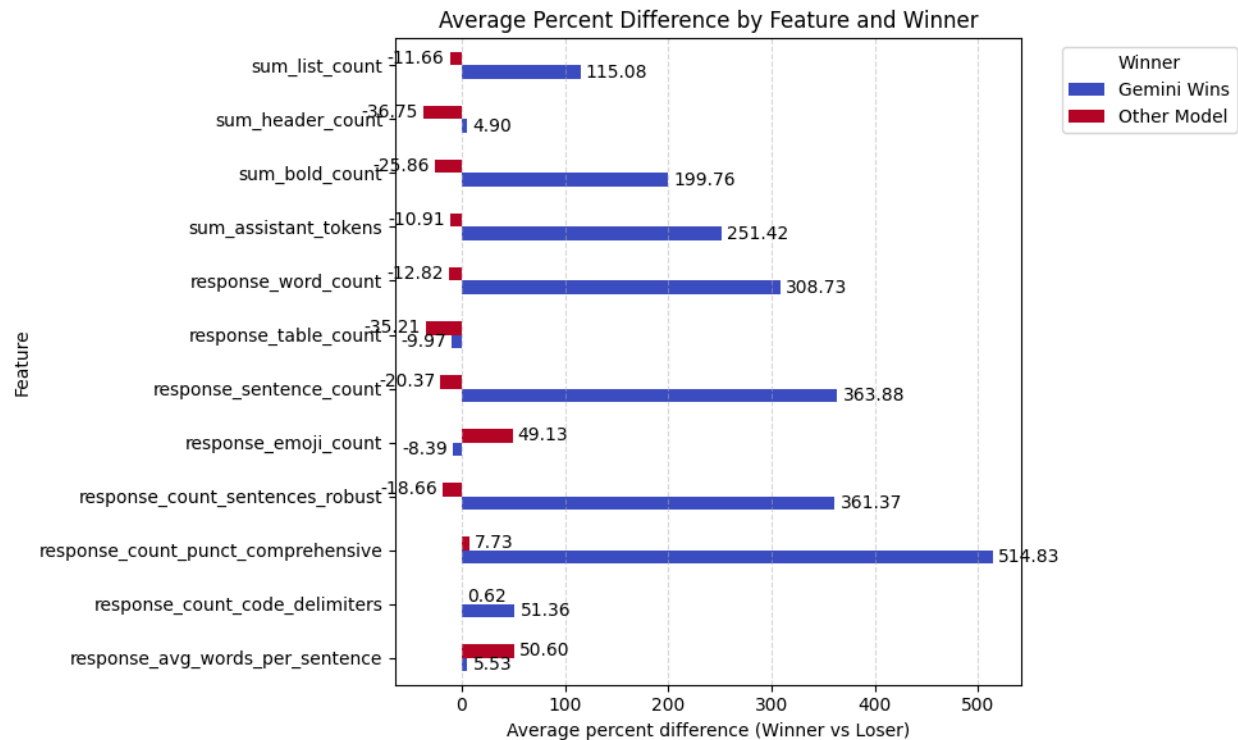


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Plot 6: Average Percent Difference Analysis

This step is similar to Step 6 (absolute difference), but now the differences are expressed as percentages relative to the loser value. Ties and both_bad matches are excluded. The results are averaged per feature and winner and visualized using a horizontal bar chart.



Plot 7: Average Percent Difference Analysis

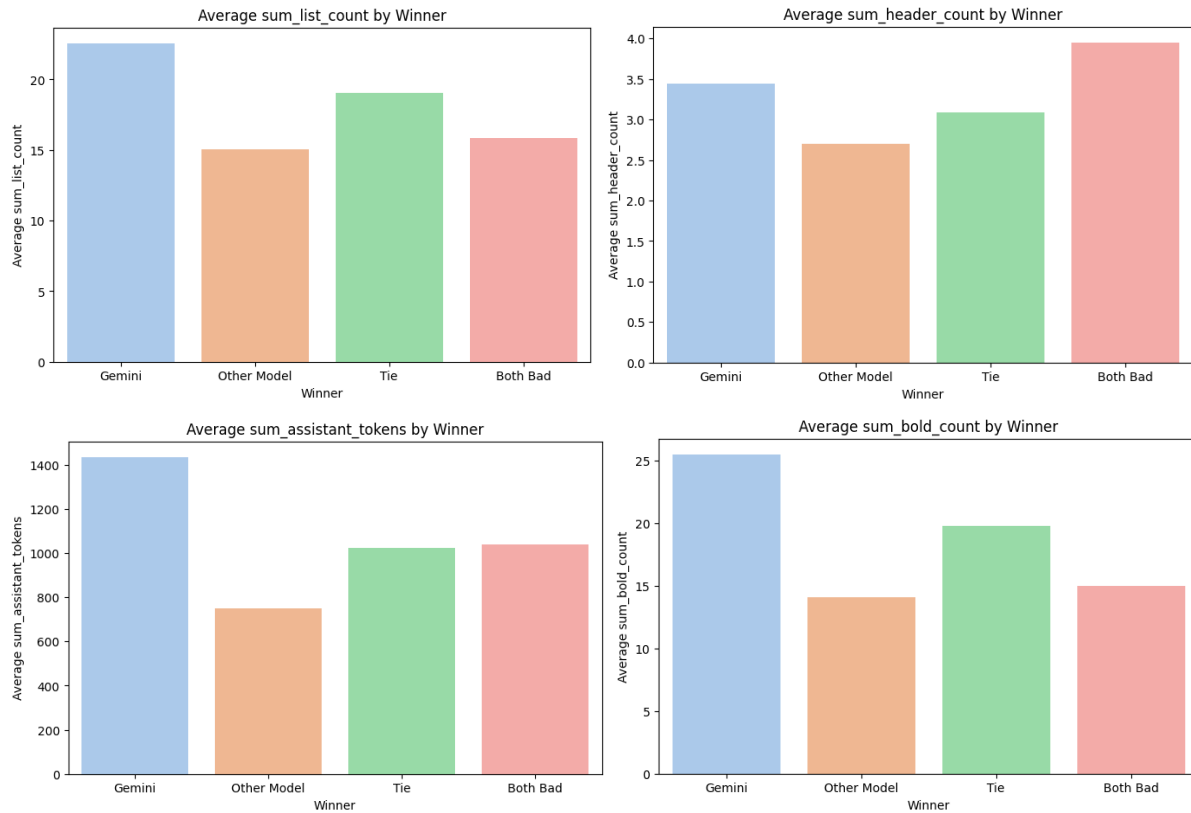
In this step, we combine feature pairs (model A and model B) into a single value depending on the winner of the match.

- If model A wins, we take the value from A.
- If model B wins, we take the value from B.
- In case of a tie, we average A and B.
- If both models are bad, we assign 0.

After combining, we calculate and visualize the **average value of each combined feature per winner** using bar plots.

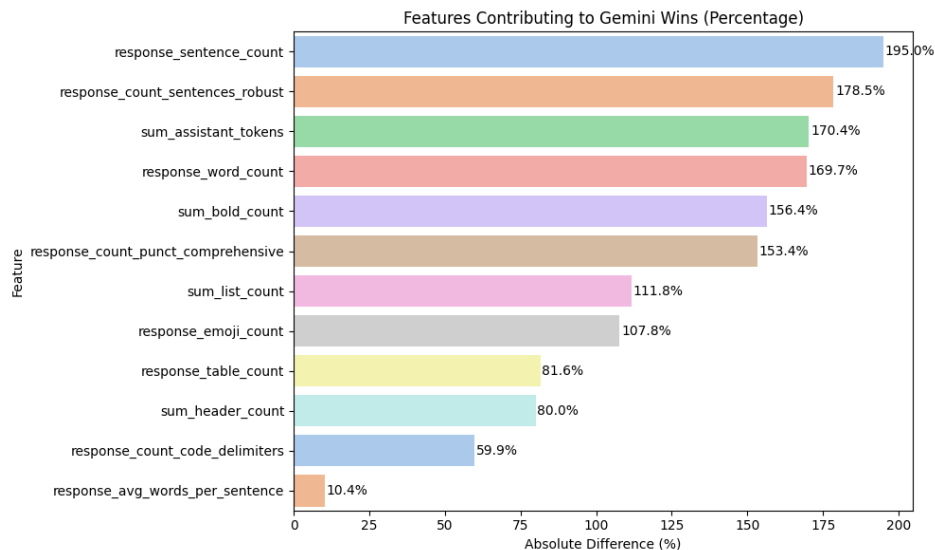
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Plot 8: Average Percent Difference Analysis

In this step, we calculate which features have the largest contribution to Gemini winning over other models. We combine feature pairs based on the winner, then compute the average value of each feature for Gemini and for other models. Finally, we calculate the absolute percentage difference and visualize the features contributing most to Gemini wins using a horizontal bar plot.



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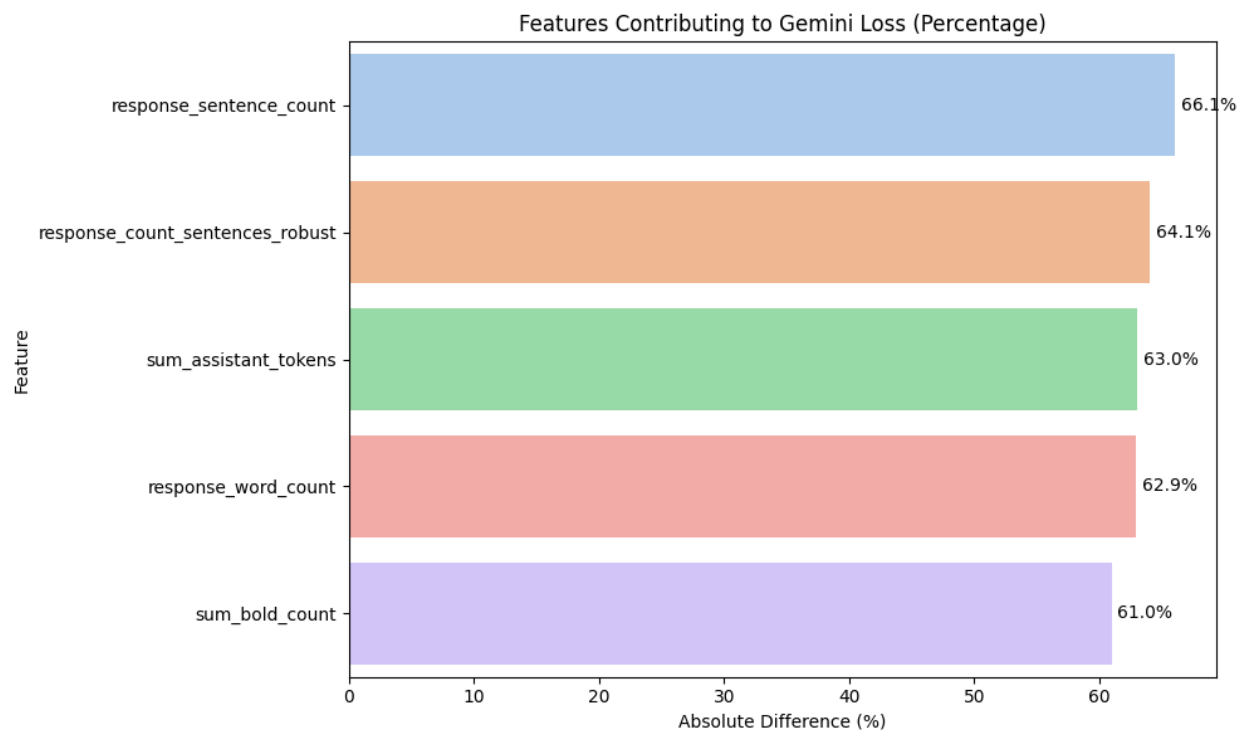
The value **170.4%** means that for this specific feature, **Gemini's average value is 170.4% higher than the average value of the other models.**

In other words, if the other models have an average value of 1 for this feature, Gemini has an average value of: 2.704. This percentage quantifies how much this feature contributes to Gemini's wins compared to the other models.

Plot 9: Features Contributing to Gemini Loss

In this step, we analyze which features are associated with Gemini losing to other models. We combine feature pairs (model A and model B) based on the winner of the match, then calculate the average value of each combined feature for matches where Gemini lost and where Gemini won.

Finally, we compute the absolute percentage difference and visualize the features that most contribute to Gemini's losses using a horizontal bar plot.



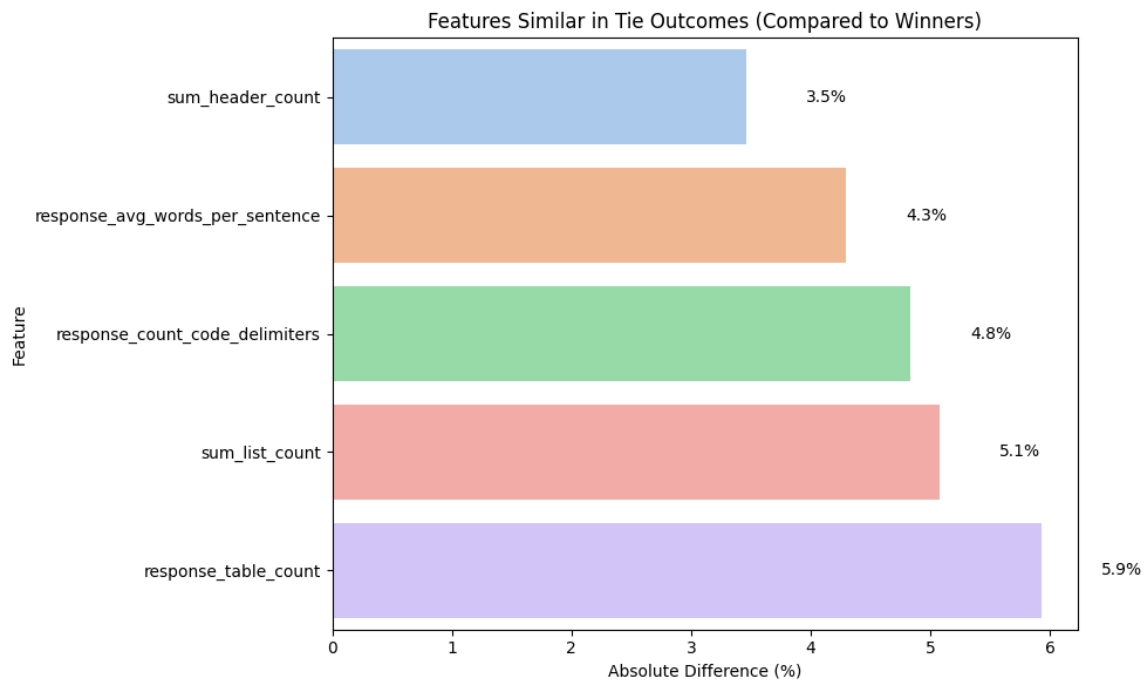
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Plot 10: Features Similar in Tie Outcomes

In this step, we identify which features have similar values in tie matches compared to matches with a clear winner. We combine feature pairs (model A and model B) based on the winner of each match.

Then, we calculate the average feature values for tie matches and for winner matches, compute the absolute percentage difference, and visualize the features that are most similar in tie outcomes using a horizontal bar plot.



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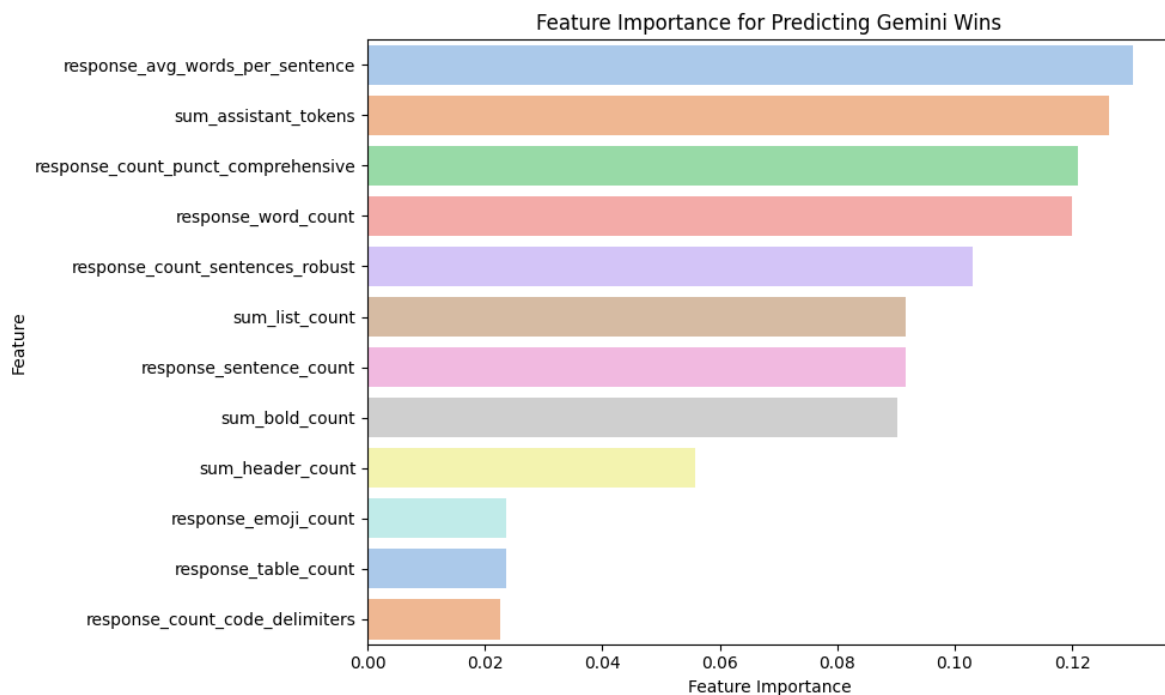
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Plot 11: Random Forest Feature Importance for Predicting Gemini Wins

In this step, we use a Random Forest classifier to identify which features are most important for predicting Gemini wins. We first remove rows where both models are bad, then combine feature pairs into a single value based on the winner.

We define the target as 1 if Gemini won (model A) and 0 otherwise. Finally, we train the Random Forest, extract feature importances, and visualize them using a horizontal bar plot.

Random Forest accuracy on test set: 0.55

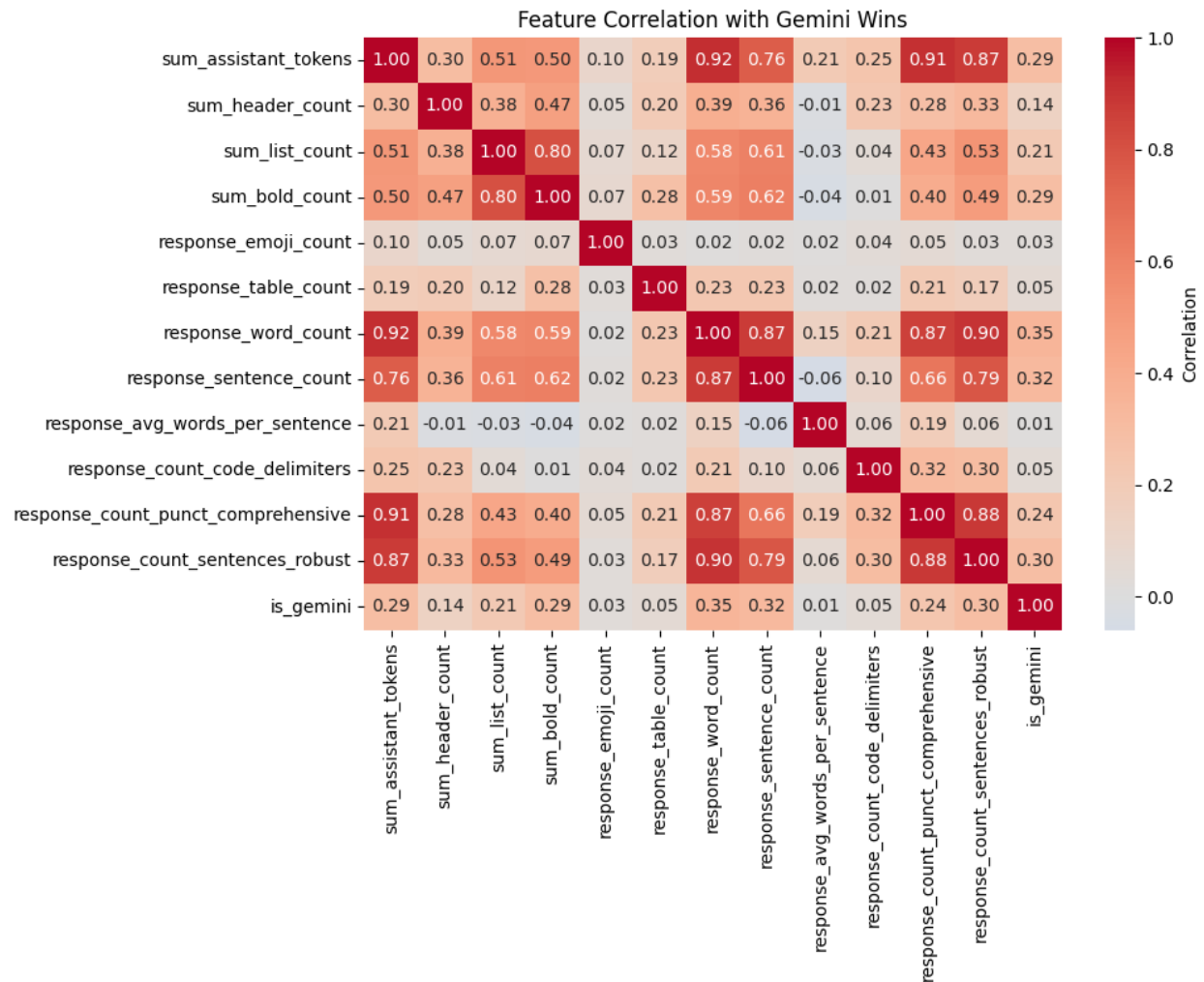


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Plot 12: Correlation Analysis with Gemini Wins

In this step, we analyze the correlation between features and Gemini winning. We create a binary target column *is_gemini* indicating whether Gemini won the match. We combine the feature pairs (A/B) into single values depending on the winner of each match. Finally, we compute the correlation matrix including the target and visualize it as a heatmap to observe which features are most associated with Gemini wins



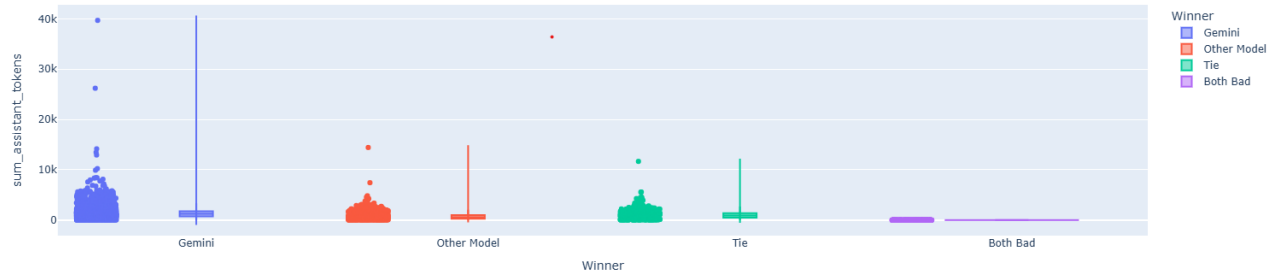
Plot 13: Correlation Analysis with Gemini Wins

In this step, we create interactive violin plots for each combined feature to visualize their distribution across different winners. We combine feature pairs (A/B) into a single column depending on the winner of the match. Winner names are mapped from numeric codes to labels (Gemini, Other Model, Tie, Both Bad). Interactive violin plots are generated for each feature, showing the full distribution, box plot, and individual points for deeper exploration.

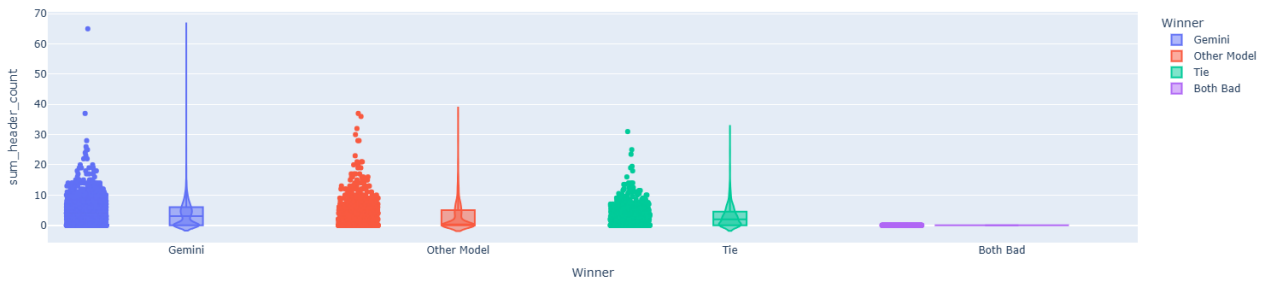
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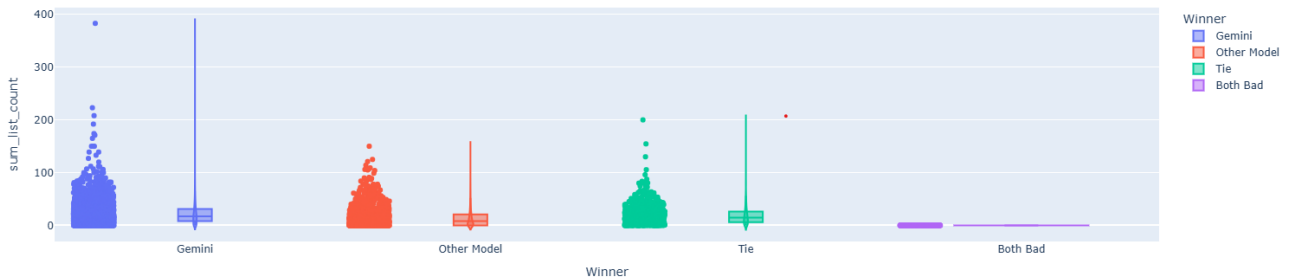
sum_assistant_tokens Distribution by Winner



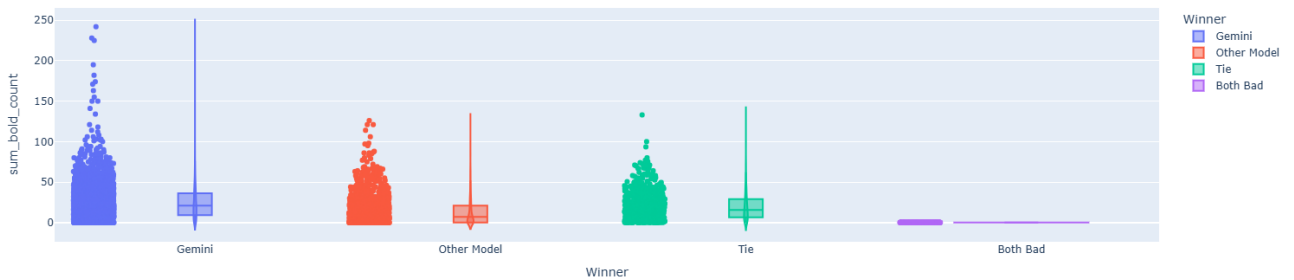
sum_header_count Distribution by Winner



sum_list_count Distribution by Winner



sum_bold_count Distribution by Winner



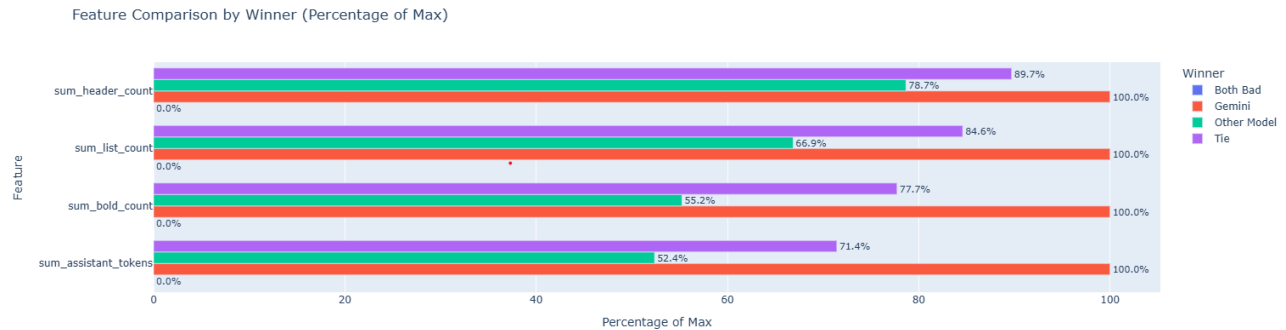
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Plot 14: Feature Comparison by Winner (Percentage of Max)

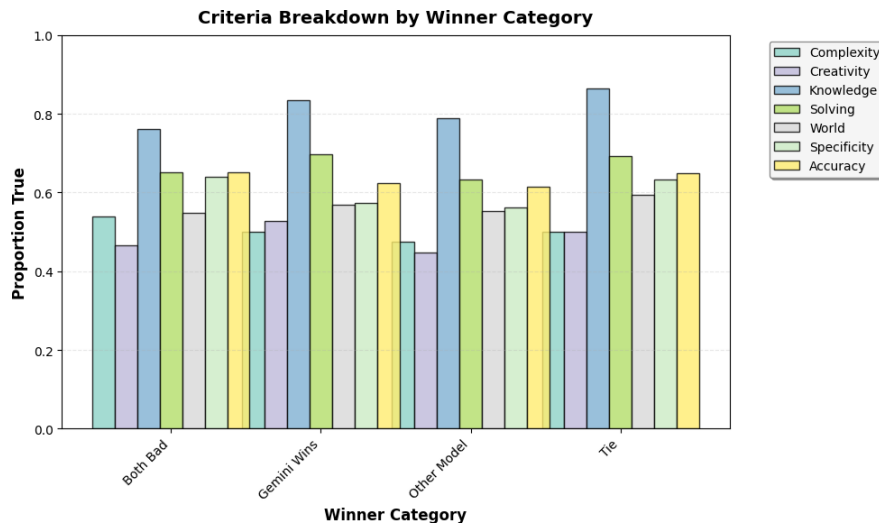
In this step, we normalize each feature per winner relative to the maximum value across all winners to express it as a percentage.

We combine feature pairs (A/B) into single columns depending on the winner. Mean values of each feature are calculated per winner. Each feature value is converted to a percentage of the maximum value for that feature. Finally, grouped bar plots are generated showing the normalized feature comparison across winners with percentages displayed on the bars.



Plot 15: Criteria Breakdown by Winner

In this step, we visualize how different evaluation criteria relate to the outcome of the matches. We focus on the *criteria_v0.1* features extracted from the dataset (e.g., *complexity*, *creativity*, *domain knowledge*, *problem solving*, *technical accuracy*). Winner codes are mapped to readable labels: Gemini, Other Model, Tie, Both Bad. For each winner category, we calculate the mean value of each criterion (proportion of True). A grouped bar chart is created, showing the relative contribution of each criterion across winner categories. This plot helps us identify which criteria are more commonly satisfied in matches won by Gemini compared to other outcomes, giving insight into what factors correlate with Gemini's success.



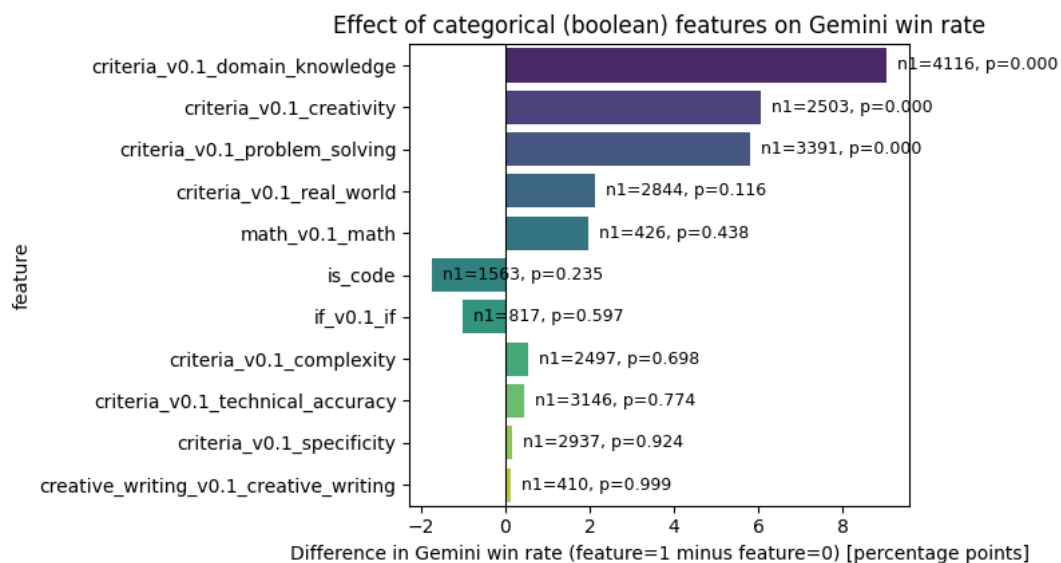
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Plot 16: Effect of Categorical (Boolean) Features on Gemini Win

In this step, we analyze how different categorical (boolean) features influence the likelihood of Gemini winning a match. We focus on boolean features extracted from the dataset (e.g., *creative_writing*, *criteria_v0.1*, *if_v0.1*, *math_v0.1*, and other 0/1 columns).

For each feature, we compute the **Gemini win rate** when the feature is 1 versus when it is 0. The difference in win rates (feature=1 minus feature=0) is calculated, indicating whether the presence of the feature favors Gemini. A horizontal bar chart is plotted showing these differences in percentage points, with annotations for sample counts and p-values from chi-squared tests. This visualization highlights which boolean features are most strongly associated with Gemini's success and provides insight into factors that increase the likelihood of winning.



feature	rate_when_1	rate_when_0	diff	pval	count_1	count_0
criteria_v0.1_domain_knowledge	0.695	0.605	0.090	0.0000	4116	891
criteria_v0.1_creativity	0.710	0.649	0.061	0.0000	2503	2504
criteria_v0.1_problem_solving	0.698	0.640	0.058	0.0000	3391	1616
criteria_v0.1_real_world	0.688	0.667	0.021	0.1160	2844	2163
math_v0.1_math	0.697	0.678	0.020	0.4384	426	4581
is_code	0.667	0.685	-0.017	0.2352	1563	3444
if_v0.1_if	0.671	0.681	-0.010	0.5974	817	4190
criteria_v0.1_complexity	0.682	0.676	0.006	0.6977	2497	2510
criteria_v0.1_technical_accuracy	0.681	0.677	0.004	0.7740	3146	1861
criteria_v0.1_specificity	0.680	0.678	0.002	0.9243	2937	2070
creative_writing_v0.1_creative_writing	0.680	0.679	0.001	0.9993	410	4597

Numeric correlations: Positive correlation means higher feature values are associated with a higher probability of Gemini winning. The p-value shows whether the correlation is statistically meaningful (e.g., $p < 0.05$ suggests the correlation is unlikely to be due to chance).

Categorical differences: (0/1 features): The value *diff* is $rate(feature=1) - rate(feature=0)$. Positive values mean the feature being 1 increases Gemini's win rate. The chi-square p-value indicates whether this difference is likely to be real rather than noise.

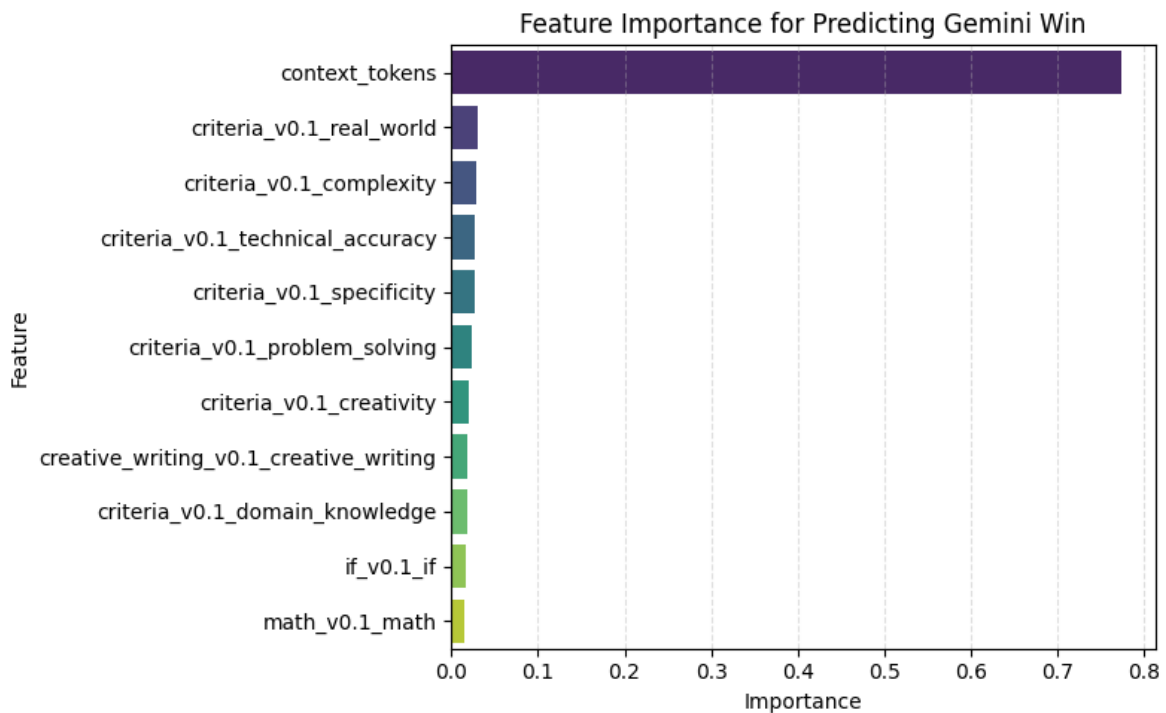
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Plot 17: Feature Importance for Predicting Gemini Win

In this step, we use a Random Forest classifier to quantify the relative importance of different features in predicting whether Gemini wins a match. We include the numeric feature *context_tokens* and categorical (boolean) features extracted from the dataset (e.g., *creative_writing*, *criteria_v0.1*, *if_v0.1*, *math_v0.1*).

The Random Forest model is trained on a stratified train-test split of the dataset with class balancing. Feature importance's are extracted from the trained model, reflecting each feature's contribution to the predictive performance. A horizontal bar chart visualizes these importance's, allowing a clear comparison of which features most strongly influence the likelihood of a Gemini win. Additionally, probability predictions and predicted labels for the test set are computed, enabling evaluation of model performance, e.g., accuracy and per-sample predicted probabilities. This analysis helps identify key factors—both numeric and boolean—that drive Gemini's success in matches.



Random Forest accuracy on test set: 0.55

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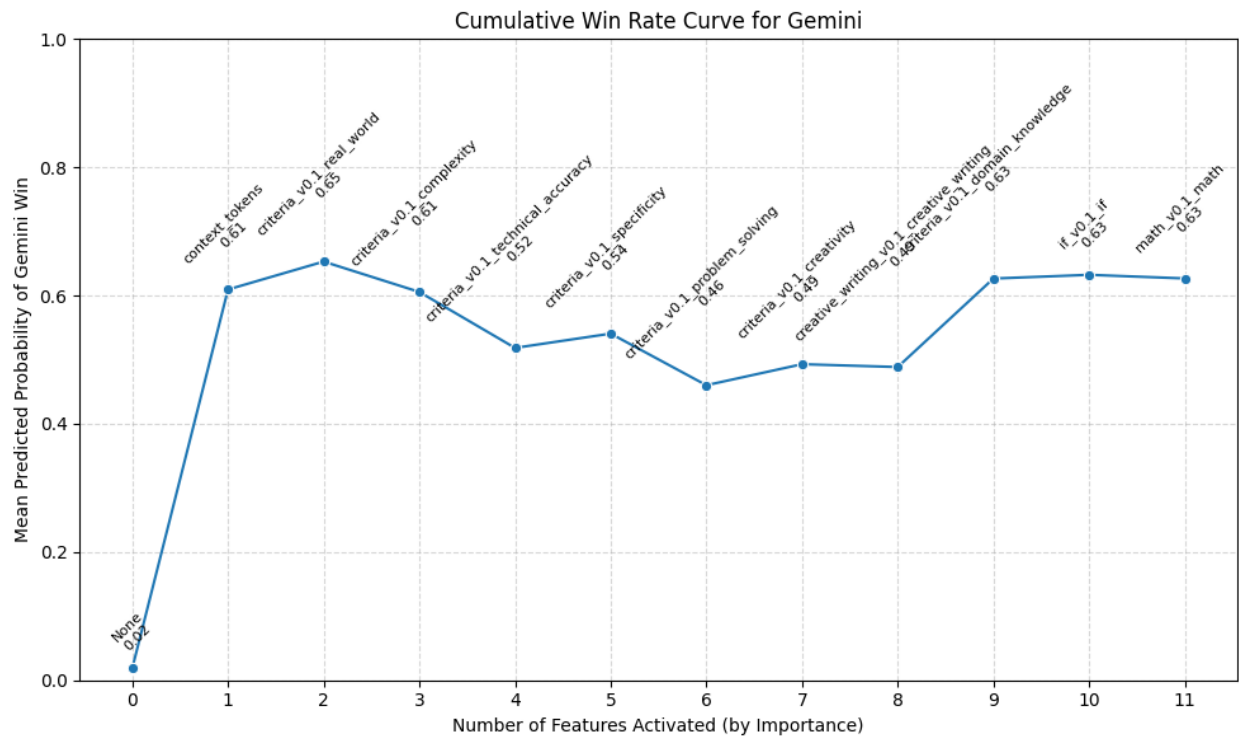
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Plot 18: Cumulative Win Rate Curve for Gemini

In this step, we visualize how the predicted probability of Gemini winning evolves as features are gradually activated according to their importance. We use the features identified previously (both numeric and boolean, including *context_tokens*, *creative_writing*, *criteria_v0.1*, *if_v0.1*, *math_v0.1*) and order them by their *Random Forest feature importance*.

Starting with all features inactive, we calculate the baseline predicted probability of a Gemini win. Then, one by one, features are activated in order of importance, and the mean predicted probability of a Gemini win across the test set is computed at each step. A cumulative line plot is drawn showing the progression of Gemini's predicted win probability as more features are included. Each point on the curve is annotated with the corresponding feature name and the predicted probability, highlighting which features contribute most to increasing the likelihood of a Gemini win.

This visualization provides an intuitive understanding of how individual features cumulatively affect model predictions and which factors are most influential in achieving a Gemini win.

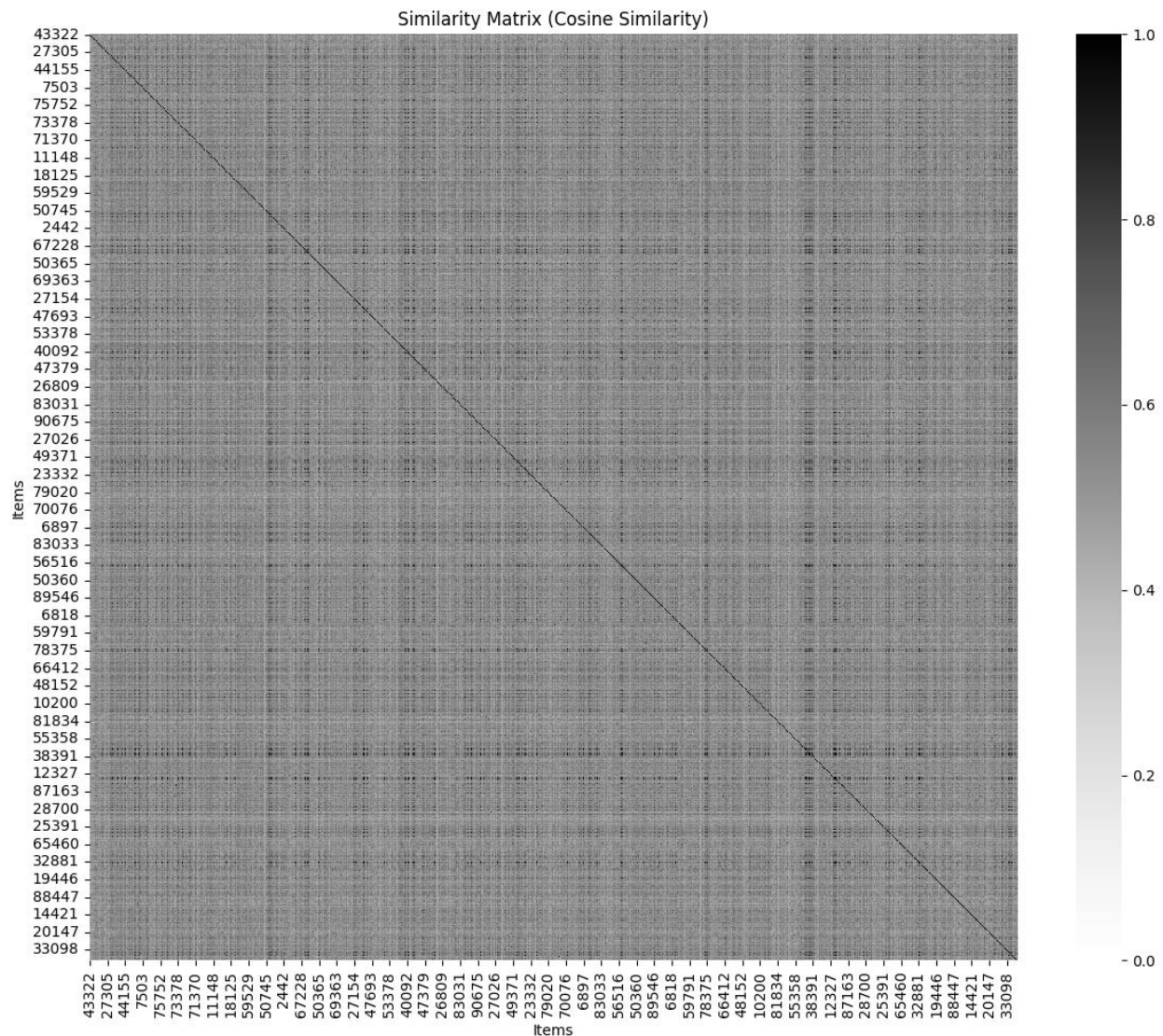


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Plot 19: Similarity Matrix (Cosine Similarity)

This figure shows a cosine similarity matrix for a random sample of 1,000 items from the dataset. Each cell in the heatmap represents the similarity between a pair of items, where lighter colors indicate lower cosine similarity values (closer to 0) and darker colors indicate higher similarity (closer to 1). The diagonal line from the top-left to the bottom-right is very dark because each item is perfectly similar to itself (similarity = 1). The generally uniform, noisy gray pattern off the diagonal suggests that most item pairs have moderate to high similarity, with a few slightly lighter or darker bands indicating big groups of items that are more similar or more dissimilar to each other compared to the rest.

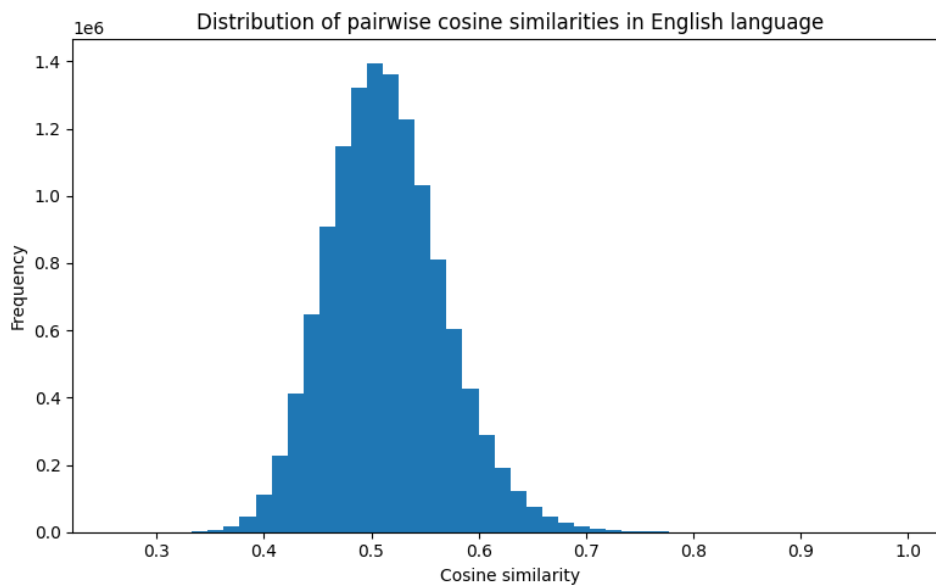
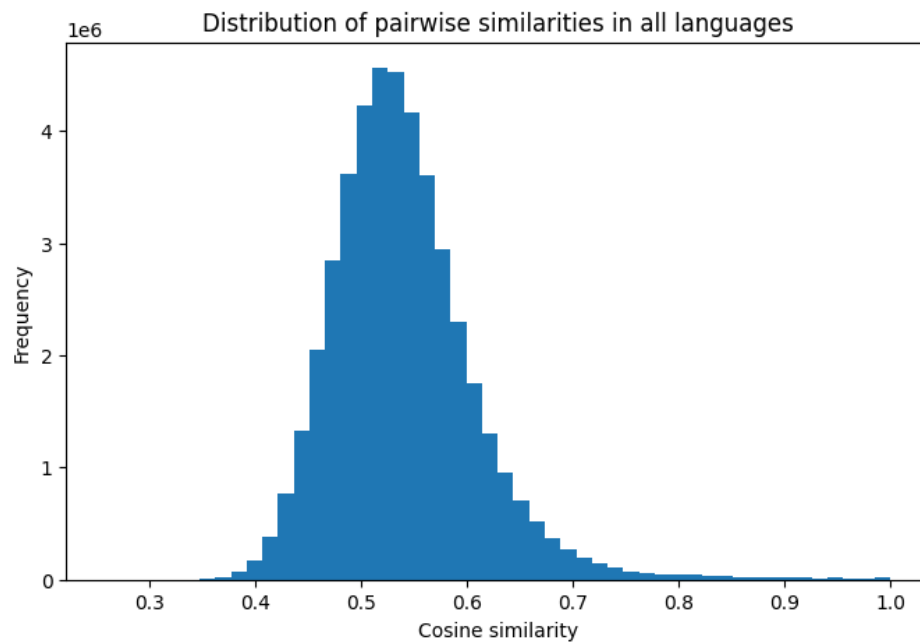


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Plot 20: Distribution of pairwise similarities

These two histograms show the distribution of pairwise cosine similarities between item embeddings. In the first plot, all languages are included: each bar represents how many item pairs fall into a given similarity range, and the distribution is roughly bell-shaped, peaking around a cosine similarity of about 0.5–0.55, with relatively few pairs below 0.35 or above 0.8. The second plot repeats the same analysis but only for items in English. This English-only distribution is also approximately normal and centered near 0.5, but appears slightly narrower, suggesting that English item embeddings are more consistently similar to each other and have fewer extremely low or high similarity pairs compared with the multilingual set.



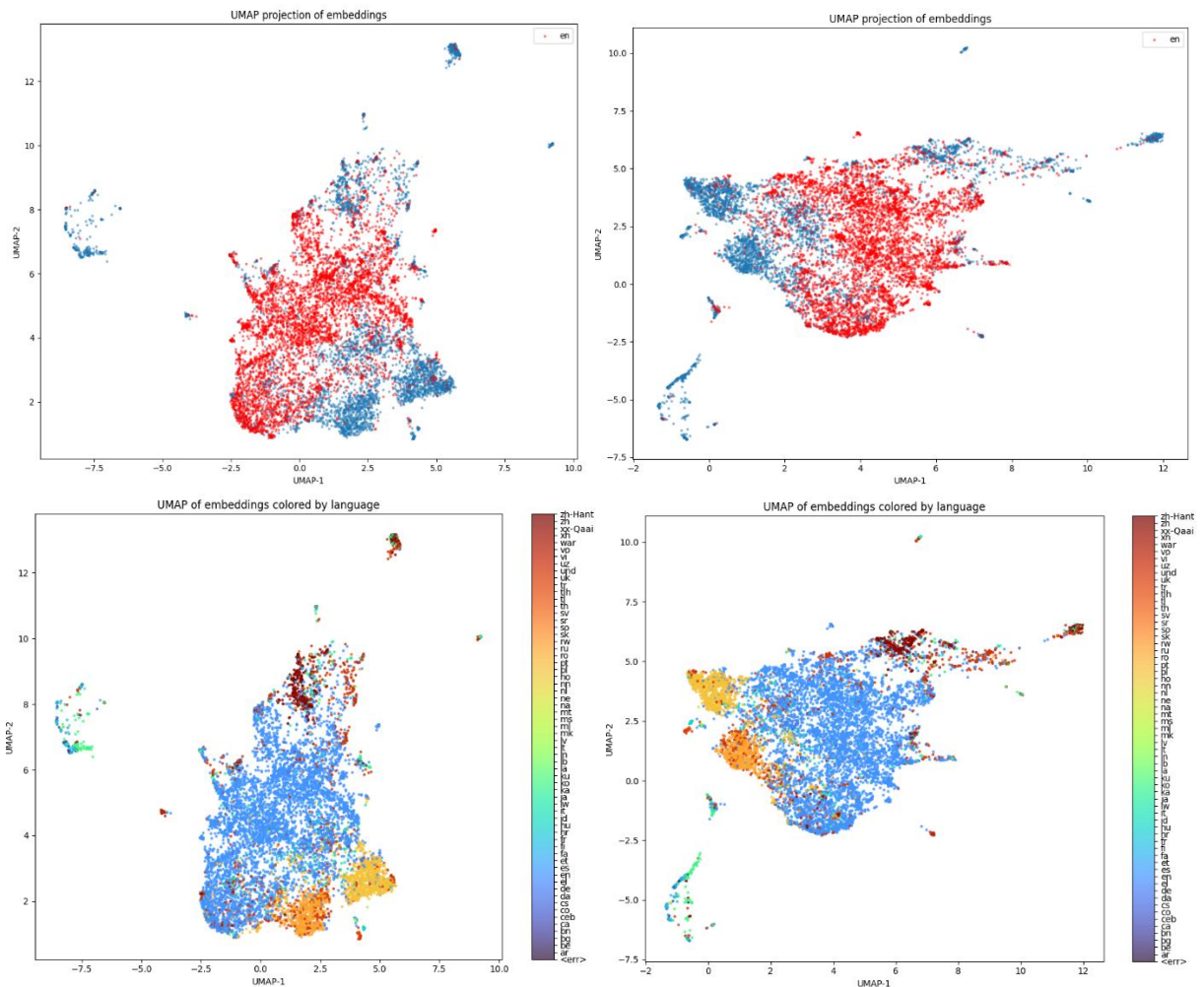
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Plot 21: UMAP projection of embeddings

These figures show a two-dimensional UMAP projection of the embedding space, where each point is an item and its position reflects its semantic relationships to other items. In the first-row plots, English items (red) are overlaid on top of all other languages (blue): the English points are spread across the main central cluster rather than forming a separate island, indicating that English and non-English items largely share the same semantic space with only a few small isolated groups.

The second-row plots colors each point by its language, revealing that many languages overlap heavily in the central region, while a few languages form more compact peripheral clusters. Overall, these visualizations suggest that the multilingual embedding model is largely language-agnostic—bringing semantically similar items from different languages close together—while still preserving some language-specific structure in certain areas of the space. The figures in the first column are drawn using the ‘Euclidean’ metric and the second column is drawn using the ‘cosine’ metric.



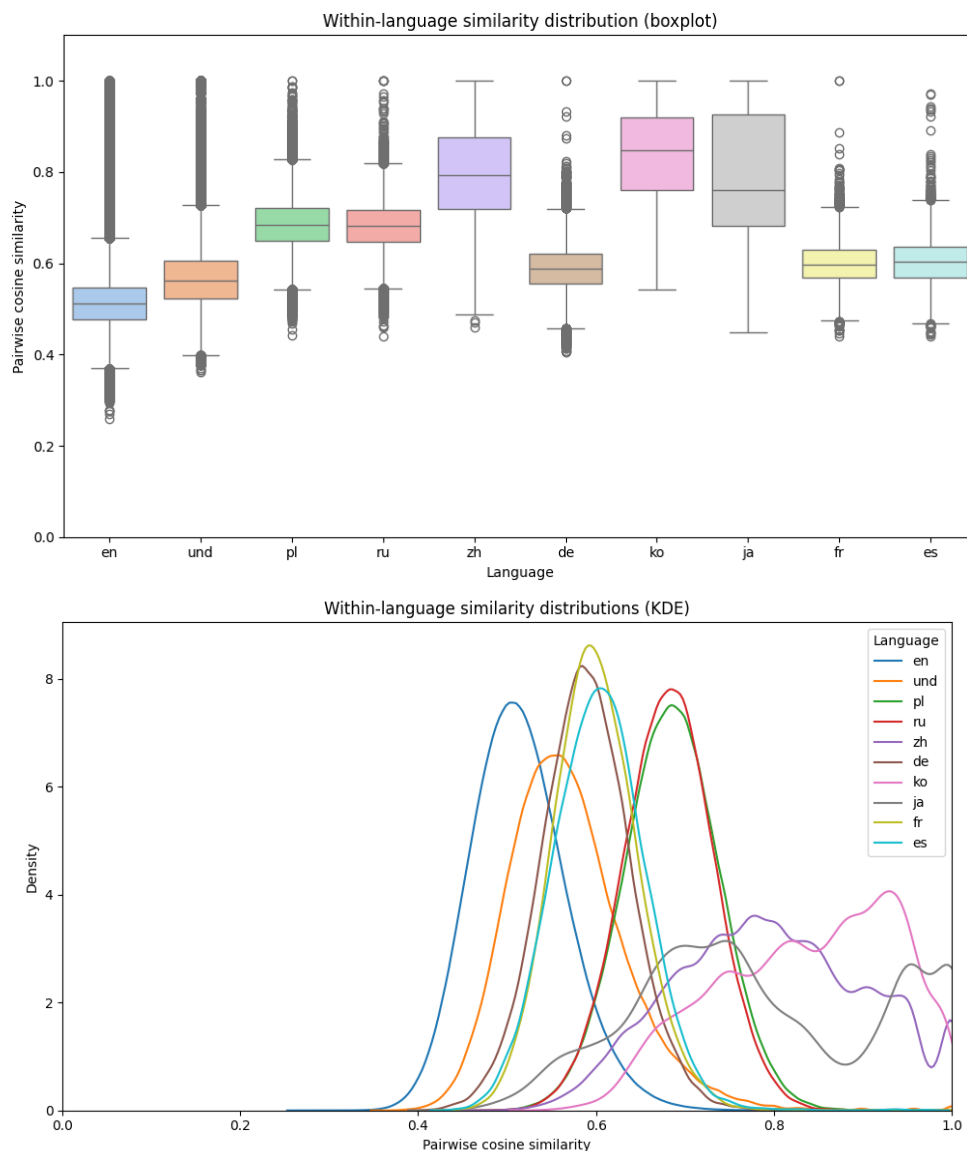
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plot 22: Within-language similarity distribution

These two plots summarize how similar items are within each language in the embedding space. The boxplot shows the distribution of pairwise cosine similarities for several languages: English has a median around 0.5, while other languages such as Polish, Russian, Chinese, German, Korean, and Japanese have noticeably higher medians and tighter interquartile ranges, indicating that their embeddings tend to be more densely clustered.

The KDE plot gives a smoother view of the same distributions and highlights how they overlap: most languages concentrate in the 0.5–0.8 similarity range, but with different peaks and spreads, and some (e.g., Chinese, Japanese, Korean) are skewed toward very high similarities. Overall, these visualizations suggest that the model organizes items in a broadly consistent way across languages, but the compactness of the embedding space varies by language, with several non-English languages exhibiting stronger within-language cohesion than English.



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Plot 23: mean cosine similarity between pairs of languages

These two figures summarize how similar the embedding space is across languages. The heatmap shows the **mean cosine similarity between pairs of languages**: values on the diagonal (within-language similarity) are highest, with especially large values for Chinese, Korean and Japanese (around 0.8+), indicating that items in these languages form very tight clusters, while English has a lower mean similarity and is more dispersed. Off-diagonal values are only slightly smaller, typically around 0.55–0.65, and are higher for related language pairs such as Polish–Russian or Chinese–Korean/Japanese, suggesting that semantically similar content in related languages lies close together in the shared space.

The similarity graph keeps only **strong connections (similarity ≥ 0.7)** for a sample of 500 items: the dense multi-colored core shows many cross-lingual high-similarity links, meaning the model often aligns semantically equivalent items across languages, while the smaller chains and isolated groups around the periphery reflect more language-specific clusters that have fewer very close counterparts in other languages.

